

Course: GISC 6328:
Drone & Remote Sensing
Terms: Summer, 2026

Meetings: Monday and Wednesday from 1:00pm - 4:45pm
Location: GR 3.206

Instructor:

- Dr. Hao Chen
- Email: Hao.Chen2@utdallas.edu
- Office Hour: Wednesday 9:00 AM -11:00 AM (with appointment)
- Office Location: GR 2.816

Teacher assistant:

- Forrest Tian
- Email: Shuang.Tian@utdallas.edu
- Office Location: GR 3.414
- Office Hour: Friday 2:00 PM -4:00 PM (with appointment)

Class Delivering:

In person: **Students meet at the scheduled class day, time and location.**

- We will also use eLearning for this class. Please contact the instructors through eLearning or MS Teams for all class related issues.

Course Description

- Recent technological advancement and Federal Aviation Administration (FAA) regulation now make it possible for a growing number of individuals and commercial firms to own small unmanned aerial systems (sUAS) - often called **drones**. This offers new opportunities for many businesses operation and scientific research, including collecting high resolution remotely sensed images, which were traditionally acquired by sensors onboard satellite and manned aircraft platforms.
- As an alternative remote sensing platform, drone provides a person with a complete control on the type of aircraft used (e.g., fixed-wing or multi-rotor drone), the sensor system (e.g., visible, NIR, and thermal camera), location, time of day, altitude above-ground-level (AGL), look-angle (e.g., oblique, vertical), look-direction, and duration, and at a reasonable cost, which was impossible before.
- To enable more students to perform personal remote sensing using drone, the class is offered at the graduate level. The course will cover the theories, technologies, methodologies, and application of drone based photography and videography for remote sensing purposes.

- The course is open not only to GIS students, but any students who are interested in drone technology at UTD.

Course Outcomes:

Upon completing this class, students will be able to

- Understanding principles of drones, remote sensing and photogrammetry
- Becoming familiar with the important technological components of different types of drones, sensors, and knowing how to build a drone yourself.
- Knowing principles of flight mission objectives and mastering mission planning software
- Operating remote controls to fly drones manually or with programmed flight lines through hands-on practice
- Processing drone photography and videography data to form orthomosaics and 3D digital surface models using structure from motion (SFM) software.
- Visualizing 3D models of drone data and performing accuracy assessment
- Performing data analysis to extract features from drone data using GIS and remote sensing software
- **Learning the safety and regulations to fly drones legally and safely, including knowledge required to obtain FAA Part 107 Remote Pilot Certificate**
- **Applying information extracted from drone photography and videography to independently solve various real-world problems.**

Pre-requisites

- **No formal pre-requisites.** Taking GICS 6325 (Remote Sensing Fundamentals) is recommended.

Textbooks

- Jensen, John R., Drone Aerial Photography and Videography: Data Collection and Image Interpretation. Kindle Edition. https://www.amazon.com/Drone-Aerial-Photography-Videography-Interpretation-ebook/dp/B07929YHLS/ref=sr_1_1?keywords=Drone+Aerial+Photography+and+Videography&qid=1552960996&s=gateway&sr=8-1

Tentative Schedule

Week Number	Topics, Reading and Lab Assignment
1	<u>Monday:</u> Introduction to the Course (Ch 1)/ Drone Anatomy (Ch2) ● Introduction ● How to use eLearning

	• Understanding basics of drone and principles of remote sensing and photogrammetry (Ch1) • Becoming familiar with the important technological components of different types of drones, sensors, and their applicability. (Ch2) • Identify parts of drone, build your own drone from parts. • 01-FAA Remote Pilot Test Preparation • Lab 1: Install, identify, search for drone/remote and other components. • FAA Quiz 01-Regulation <u>Wednesday: FAA Legal Consideration (Ch3)</u> • Learning the safety and regulations to fly drones legally and safely, including knowledge required to obtain FAA Part 107 Remote Pilot Certificate (Ch3). • 02-FAA Remote Pilot Test Preparation • Quiz 1
2	<u>Monday: Manual Flying</u> • Operating remote controls to fly drones (DJI) manually or with programmed flight lines through hands-on practice • Lab 2: Manual flying drones (such as DJI Phantom 4, Mavic 2/3) to achieve goals such as taking off, moving forward and backward, turning left and right, returning home, landing and collecting still images. • 03-FAA Remote Pilot Test Preparation • FAA Quiz 02-Weather <u>Wednesday: Collect Aerial Photography and Videography (Ch 4)</u> • Knowing principles of flight mission planning and collecting drone photography and videography (Ch 4) • 04-FAA Remote Pilot Test Preparation 04 • Quiz 2
3	<u>Monday: Mission Planning and Autonomous Flying</u> • Mastering mission planning software: such as Drone Deploy, DJI Go, Mission planner. • 05-FAA Remote Pilot Test Preparation 05

	• Lab 3: Plan Mission. Developing a data collection plan with mission planning software; Automatic Flying and Data Collection with Mission Plan (DJI M3M). • FAA Quiz 03 workload and emergency <u>Wednesday:</u> • Midterm Examination
4	<u>Monday: Structure from Motion</u> • Processing drone photography and videography data to form Ortho mosaics and 3D digital surface models • ArcGIS Drone2Map Introduction • 06-FAA Remote Pilot Test Preparation • Lab 4: Use Drone2Map to create 2/3D products • Quiz 3 <u>Wednesday: Interpreting Arial Photography and Videography (Ch5)</u> • Visualizing 3D models and performing data analysis to extract features from drone data using GIS and remote sensing software (Ch5) • 07-Become a FAA Certified Pilot • Final project discussion • Quiz 4
5	<u>Monday: Applications (Ch 6, 8, 9)</u> • Applying information extracted from drone photography and videography to different applications, such as agriculture, forestry, archaeology, construction site inspection, disaster assessment and emergency response, habitat assessment and animal population estimation, geospatial intelligence, law enforcement, search & rescue, firefighting, mine, landfill, and recycling monitoring, real estate, roof and chimney inspection, transportation infrastructure and utility facility monitoring. (Ch 6) • Learning to become a FAA Certified Remote Pilot (Ch 7, 8) • 08-FAA Remote Pilot Sample Test • Lab 5: Object Detection and Segmentation using Deep Learning with Pretrained Models • FAA Quiz 04 Sectional Chart and Operation

	<u>Wednesday: Individual Final Project Presentation (no class)</u> • All students work on the final project with a topic of interest they identified. • Graduate students will take a Mock Remote Pilot Exam • Present a project using drone data for an application of interest. Submit a link to video recording of your presentation (10 minutes)
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Grading Policy

- Graduate students: Pop Quiz (10%), Labs (50%), Midterm Exam (10%), Mock FAA Exam (10%) and Final Project (20%)

Letter Grade	Undergraduate	Graduate	GPA
A+	>97		4
A	93 - <97	93 - 100	4
A-	90 - <93	90 - <93	3.7
B+	87 - <90	87 - <90	3.3
B	83 - <87	83 - <87	3
B-	80 - <83	80 - <83	2.7
C+	77 - <80	77 - <80	2.3
C	73 - <77	60 - <77	2
C-	70 - <73		1.7
D+	67 - <70		1.330
D	63 - <67		1
D-	60 - < 63		0.67
F	<60	<60	0

Make-up Exams

- No Make-up exam or project will be given without a legitimate reason accompanied by proper documentation (e.g., a doctor's excuse).

Late Work

- Late submission for labs will be penalized 10% per day late.

Class Attendance

- Students are expected to attend and actively participate in class discussion.

Classroom Citizenship

- Please make sure you turn off your cell-phone before coming to the classroom. Viewing anything that is not related to class and communicating with others using instant messenger are prohibited during the class.

Academic Dishonesty

- All suspected cases of academic dishonesty (cheating, plagiarism, collusion, etc.) will be immediately forwarded to the Office of Judicial Affairs. Students have a right to due process through Judicial Affairs, but if the accused student is found guilty of misconduct, Judicial Affairs has sole authority for determining punishment.

UT Dallas Syllabus Policies

- <http://coursebook.utdallas.edu/syllabus-policies/>
- These descriptions and timelines are subject to change at the discretion of the Instructor.